INDIAN STOCK MARKET ANALYSIS USING MICROSOFT AZURE

Abstract

This study presents a comprehensive approach to analyzing the Indian stock market using Microsoft Azure cloud services. The primary objective is to leverage Azure's scalable data processing, machine learning, and visualization tools to gain insights into stock price movements, trends, and predictive patterns. The Indian stock market, consisting of major exchanges such as NSE and BSE, generates massive volumes of structured and unstructured financial data, making cloud-based analytics a suitable solution for efficient processing and analysis.

The proposed system utilizes Azure Data Factory for data ingestion from financial APIs and historical datasets, Azure Blob Storage for scalable data storage, and Azure Databricks for big data processing and transformation. Machine learning models are developed using Azure Machine Learning services to perform tasks such as price prediction, trend analysis, and anomaly detection. Techniques such as Linear Regression, Time Series Forecasting, and K-Nearest Neighbors (KNN) are applied to evaluate stock performance and predict future trends.

Additionally, Azure Synapse Analytics is used for data warehousing and querying large datasets, while Power BI is integrated for interactive visualization and reporting. These visual dashboards help investors and analysts make informed decisions based on real-time insights and historical trends.

The system ensures scalability, reliability, and cost-effectiveness by utilizing Azure's cloud infrastructure. It also addresses challenges such as data latency, market volatility, and model accuracy. The results demonstrate that cloud-based stock market analysis significantly enhances prediction accuracy and decision-making efficiency compared to traditional methods.